

Volume IV · Chapter 19 — Figures & Multimodal Understanding · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-IV-Chapter-19-context.md` for the AI interaction log.

Prerequisites: Ch 18, Ch 7.

Learning outcomes — the student can: detect/extract figures; generate & curate captions with multimodal models; link images to text; make images retrievable; describe the SOTA and limits of medical image understanding.

Topics (theory) — 8 h:

#	Topic	h
19.1	Medical figures: radiology, histology, ECG, diagrams, charts	1
19.2	Figure detection & extraction from documents	1
19.3	Multimodal models (captioning, VQA): LLaVA-Med, MedGemma-multimodal	2
19.4	Image embeddings & multimodal retrieval	2
19.5	Linking figures to surrounding text & captions	1
19.6	SOTA & limits: reliability of medical image understanding	1

Datasets/tools: multimodal HF models (LLaVA-Med/MedGemma), CLIP-style embeddings; vector DB. **Assessment:** multimodal ingest+retrieval (**60%**); caption-quality report (**20%**); quiz (**20%**). **Key decisions:** which multimodal model; how much to trust captions; image vs. caption embeddings. **References:** plan §6; §13 → *Medical LLMs*. **Hours:** Theory **8** + Lab **14** = **22**.